

Direct device access from the SmartNIC towards datacenter disaggregation

Master's thesis meeting : week 1

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Table of contents

- Bdevperf benchmarks
- Basic file transfer
- Blobstore
- Conclusion
- TODOs for week 1

Bdevperf benchmarks

- Goal : match performances with my own implementation
- Sketchy screenshot for this week, proper python measurement skip later
- Focusing on the C code

Direct device access from the SmartNIC towards datacenter disaggregation (Nicolas Jeanmenne)

```
[2026-02-06 11:10:59.140201] reactor.c: 990:reactor_run: *NOTICE*: Reactor started on core 0
Running I/O for 15 seconds...
 98151.07 IOPS, 6134.44 MiB/s

Device Information      : runtime(s)      IOPS      MiB/s      Fail/s      T0/s      Average      Latency(us)      min      max
Job: Nvme0n1 (Core Mask 0x1, workload: write, depth: 64, IO size: 65536)
   Nvme0n1              :      15.00    98150.72    6134.42      0.00      0.00      652.19         83.36    17395.36
=====
Total                   :      98150.72    6134.42      0.00      0.00      652.19         83.36    17395.36
hjeanmenne@frodo:~/nvme-of/spdk$ sudo ./build/examples/bdevperf -w write -q 32 -o 65536 -t 15 --config simple-config.json
[2026-02-06 11:20:57.322201] Starting SPDK v26.01-pre git sha1 3266b7702 / DPDK 25.07.0 initialization...
[2026-02-06 11:20:57.322284] [ DPDK EAL parameters: bdevperf --no-shconf -c 0x1 --huge-unlink --no-telemetry --log-level=lib
-file-prefix=spdk_pid1047318 ]
EAL: '-c <coremask>' option is deprecated, and will be removed in a future release
EAL: Use '-l <corelist>' or '--lcores=<corelist>' option instead
[2026-02-06 11:20:57.390250] app.c: 970:spdk_app_start: *NOTICE*: Total cores available: 1
[2026-02-06 11:20:57.396598] reactor.c: 996:reactor_run: *NOTICE*: Reactor started on core 0
Running I/O for 15 seconds...
 96410.80 IOPS, 6025.68 MiB/s

Device Information      : runtime(s)      IOPS      MiB/s      Fail/s      T0/s      Average      Latency(us)      min      max
Job: Nvme0n1 (Core Mask 0x1, workload: write, depth: 32, IO size: 65536)
   Nvme0n1              :      15.00    96410.94    6025.68      0.00      0.00      332.02         81.38    13451.06
=====
Total                   :      96410.94    6025.68      0.00      0.00      332.02         81.38    13451.06
hjeanmenne@frodo:~/nvme-of/spdk$ sudo ./build/examples/bdevperf -w write -q 16 -o 65536 -t 15 --config simple-config.json
[2026-02-06 11:21:40.769189] Starting SPDK v26.01-pre git sha1 3266b7702 / DPDK 25.07.0 initialization...
[2026-02-06 11:21:40.769281] [ DPDK EAL parameters: bdevperf --no-shconf -c 0x1 --huge-unlink --no-telemetry --log-level=lib
-file-prefix=spdk_pid1047329 ]
EAL: '-c <coremask>' option is deprecated, and will be removed in a future release
EAL: Use '-l <corelist>' or '--lcores=<corelist>' option instead
[2026-02-06 11:21:40.837593] app.c: 970:spdk_app_start: *NOTICE*: Total cores available: 1
[2026-02-06 11:21:40.843921] reactor.c: 996:reactor_run: *NOTICE*: Reactor started on core 0
Running I/O for 15 seconds...
 97243.53 IOPS, 6077.72 MiB/s

Device Information      : runtime(s)      IOPS      MiB/s      Fail/s      T0/s      Average      Latency(us)      min      max
Job: Nvme0n1 (Core Mask 0x1, workload: write, depth: 16, IO size: 65536)
   Nvme0n1              :      15.00    97243.60    6077.72      0.00      0.00      164.61         21.63    13046.52
=====
Total                   :      97243.60    6077.72      0.00      0.00      164.61         21.63    13046.52
hjeanmenne@frodo:~/nvme-of/spdk$ sudo ./build/examples/bdevperf -w write -q 16 -o 4096 -t 15 --config simple-config.json
[2026-02-06 11:22:03.995070] Starting SPDK v26.01-pre git sha1 3266b7702 / DPDK 25.07.0 initialization...
[2026-02-06 11:22:03.995153] [ DPDK EAL parameters: bdevperf --no-shconf -c 0x1 --huge-unlink --no-telemetry --log-level=lib
-file-prefix=spdk_pid1047340 ]
EAL: '-c <coremask>' option is deprecated, and will be removed in a future release
EAL: Use '-l <corelist>' or '--lcores=<corelist>' option instead
[2026-02-06 11:22:04.063775] app.c: 970:spdk_app_start: *NOTICE*: Total cores available: 1
[2026-02-06 11:22:04.070187] reactor.c: 996:reactor_run: *NOTICE*: Reactor started on core 0
Running I/O for 15 seconds...
 1480538.40 IOPS, 5783.35 MiB/s

Device Information      : runtime(s)      IOPS      MiB/s      Fail/s      T0/s      Average      Latency(us)      min      max
Job: Nvme0n1 (Core Mask 0x1, workload: write, depth: 16, IO size: 4096)
   Nvme0n1              :      15.00   1480538.25    5783.35      0.00      0.00      10.76          5.31    12136.30
=====
Total                   :      1480538.25    5783.35      0.00      0.00      10.76          5.31    12136.30
hjeanmenne@frodo:~/nvme-of/spdk$ sudo ./build/examples/bdevperf -w write -q 32 -o 4096 -t 15 --config simple-config.json
[2026-02-06 11:23:19.750175] Starting SPDK v26.01-pre git sha1 3266b7702 / DPDK 25.07.0 initialization...
[2026-02-06 11:23:19.750262] [ DPDK EAL parameters: bdevperf --no-shconf -c 0x1 --huge-unlink --no-telemetry --log-level=lib
-file-prefix=spdk_pid1047351 ]
EAL: '-c <coremask>' option is deprecated, and will be removed in a future release
EAL: Use '-l <corelist>' or '--lcores=<corelist>' option instead
[2026-02-06 11:23:19.819461] app.c: 970:spdk_app_start: *NOTICE*: Total cores available: 1
[2026-02-06 11:23:19.825837] reactor.c: 996:reactor_run: *NOTICE*: Reactor started on core 0
Running I/O for 15 seconds...
 1483876.13 IOPS, 5796.39 MiB/s

Device Information      : runtime(s)      IOPS      MiB/s      Fail/s      T0/s      Average      Latency(us)      min      max
Job: Nvme0n1 (Core Mask 0x1, workload: write, depth: 32, IO size: 4096)
   Nvme0n1              :      15.00   1483876.22    5796.39      0.00      0.00      21.52          7.21    12136.30
=====
Total                   :      1483876.22    5796.39      0.00      0.00      21.52          7.21    12136.30
```

Basic file transfer

- Writing code on the bdev layer
- Allow one file to be stored into the SSD
- My estimation : code will be functional for next meeting

Blobstore

- The big next step
- Goal : implement a filesystem with SPDK blobstore
- Will be useful for file synchronization

Fosdem side quests

- Talks about hardware offloading
- One interesting about zero-copy optimization
- Discovered CephFS there but idk if it's useful

Conclusion

- Gained some experience with SPDK ecosystem
- Results ASAP

TODOs for week 1

- Demo for next week working

That's all for today !